INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Lab 3b: Advanced Cybersecurity Risk Quantification

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C11/26/CGRCE/17569

COHORT 11

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Phase 1: Risk Exposure Calculation

Task 1: Calculate Annualized Loss Expectancy (ALE)

For each vulnerability, calculate:

- Single Loss Expectancy (SLE) = Asset value x Exposure Factor
- Annual Rate of Occurrence (ARO) = Number of occurances in a year
- Annualized Loss Expectancy (ALE) = Single Loss Expectancy × Annual Rate of Occurrence

- Single Loss Expectancy (SLE) = Asset value x Exposure Factor

Vulnerability 1- API Authentication Bypass

- SLE = $5,000,000

Vulnerability 2- Database Injection

- 500,000 x $250 = $125,000,000

Vulnerability 3- Session Hijacking

- 5,000 x 1,500 = $7,500,000

- Annual Rate of Occurrence (ARO) = Number of occurances in a year

- Vulnerability 1- ARO = 15% = 0.15
- Vulnerability 2- ARO = 25% = 0.25
- Vulnerability 3- ARO = 40% = 0.40

- Annualized Loss Expectancy (ALE) = Single Loss Expectancy × Annual Rate of Occurrence
  - Vulnerability 1 = 5,000,000 x 0.15 = $750,000
  - Vulnerability 2 = $125,000,000 x 0.25 = $31,250,000
  - Vulnerability 3 = $7,500,000 x 0.4 = $3,000,000

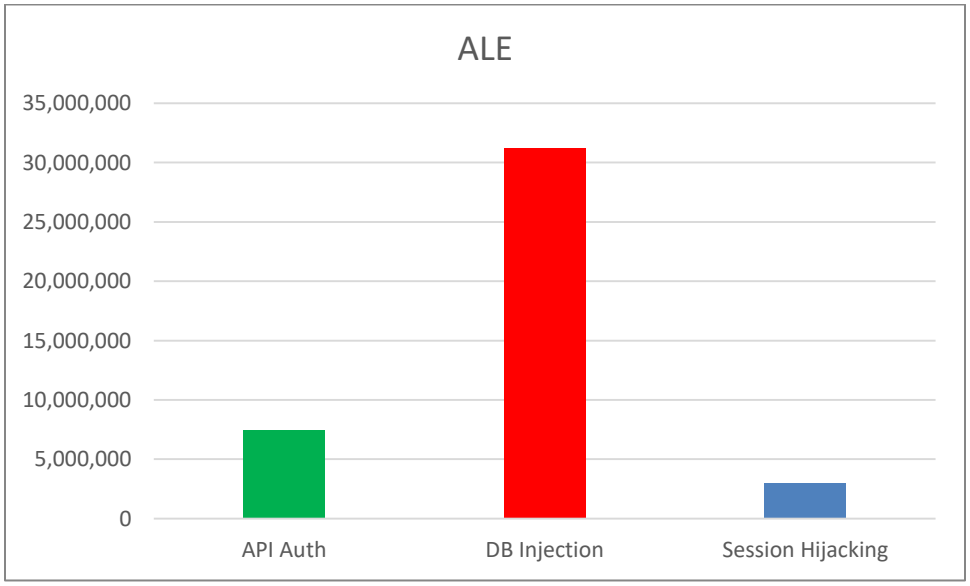
Total ALE = $35,000,000.

Task 2: Prioritize Risks

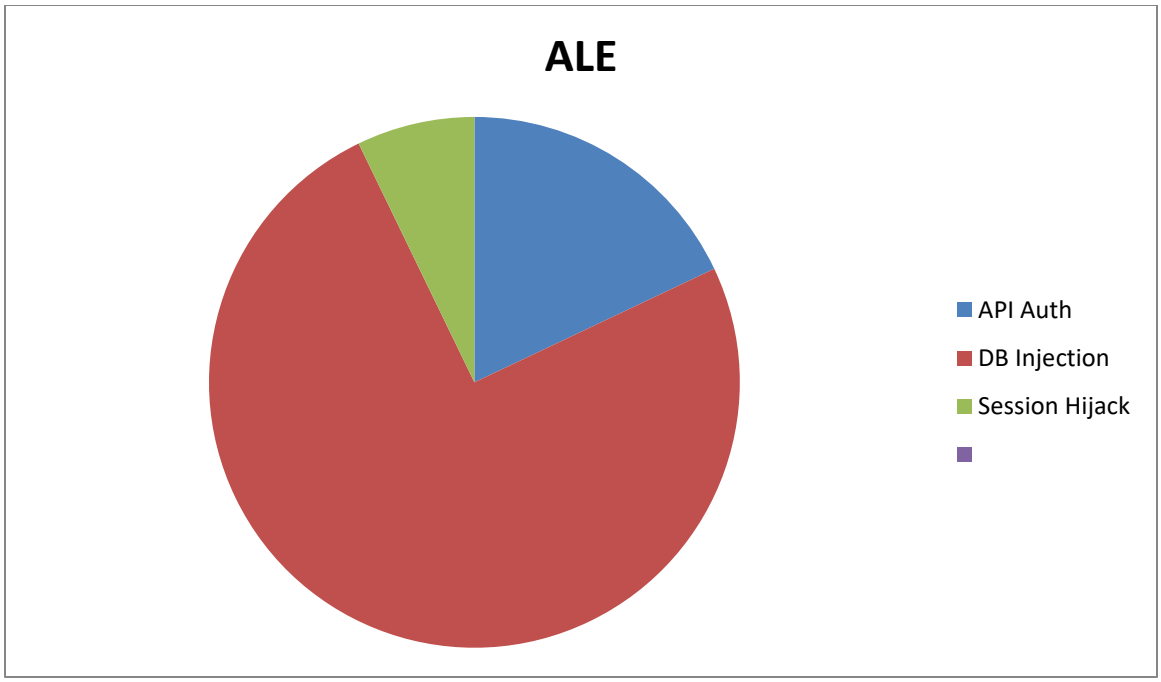
Rank	Findings	ARO	SLE	ALE	Risk Level
1	Database Injection	25%	$125,000,000	$31,250,000	Critical
2	Session Hijacking	40%	$7,500,000	$3,000,000	High
3	API Authentication	15%	$5,000,000	$750,000	High

Task 3: Create Risk Visualization

- A bar chart comparing ALE for all vulnerabilities



- A pie chart showing risk distribution



Phase 2: Control Evaluation

Task 4: Cost-Benefit Analysis

Control 1 :Advanced API Security Gateway

- Initial Investment = $350,000
- Annual Maintenance = $50,000 per year
- Risk Reduction (in $) = ALE x Effectiveness = $31,250,000 × 90% = $675,000
- Annual Net Benefit= Risk Reduction – Maintenance = $675,000– $50,000 = $625,000
- ROI (Year 1) = (Benefit ÷ Cost) × 100= ($625,000 ÷ $350,000) × 100 = 178.6%
- Payback Period = Cost ÷ Benefit= $350,000 ÷ $625,000 ≈ 0.56 years

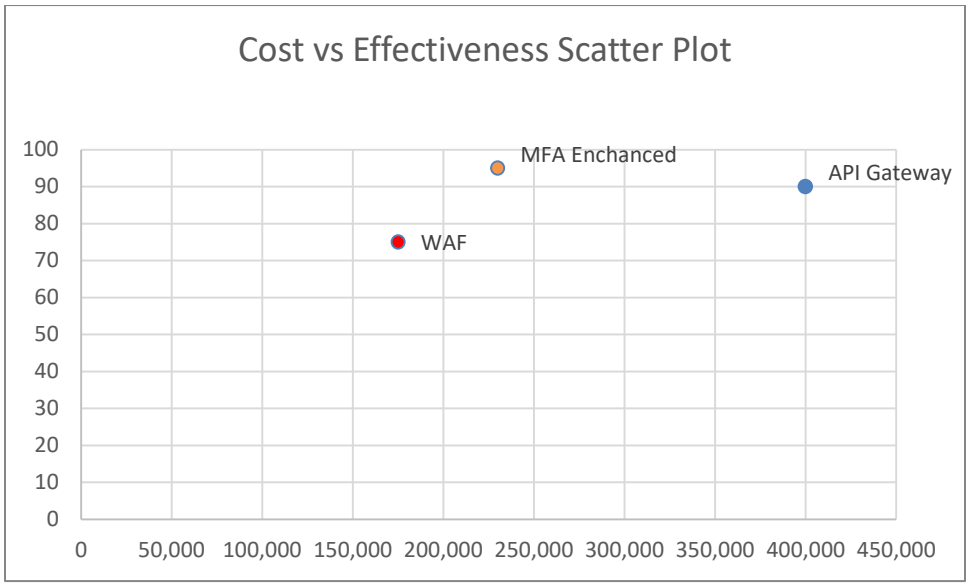
Control 2: Web Application Firewall (WAF)

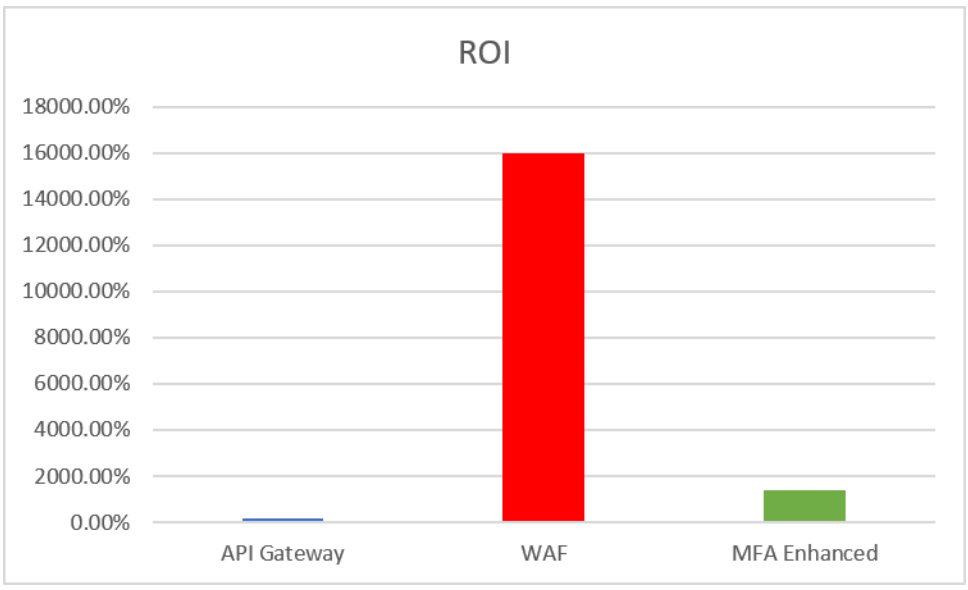
- Risk Reduction = 0.75 × ($750,000 + $31,250,000)= 0.75 × $32,000,000 = $24,000,000
- Annual Net Benefit = $24,000,000 – $25,000 = $23,975,000
- ROI (Year 1) = ($23,975,000 ÷ $150,000) × 100 ≈ 15,983%
- Payback Period = $150,000 ÷ $23,975,000 ≈ 0.006 years

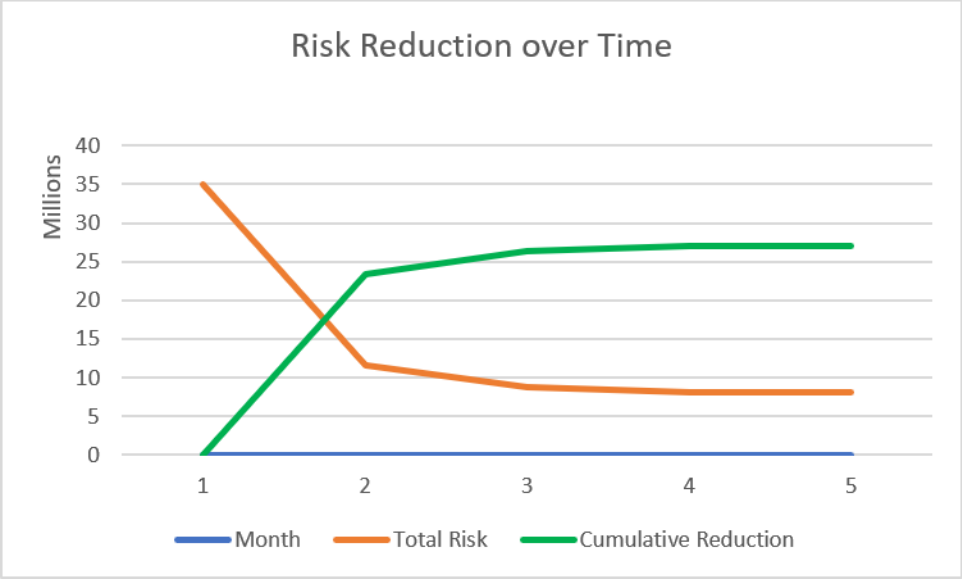
Control 3: Multi-Factor Authentication Enhancement

- Risk Reduction = 0.95 × $3,000,000 = $2,850,000
- Annual Net Benefit = $2,850,000 – $30,000 = $2,820,000
- ROI (Year 1) = ($2,820,000 ÷ $200,000) × 100 = 1,410%
- Payback Period = $200,000 ÷ $2,820,000 ≈ 0.07 years

Task 5: Control Selection Analysis







Phase 3: Executive Reporting

Task 6: Create Executive Dashboard

To: CISO and CFO, SecureBank Financial

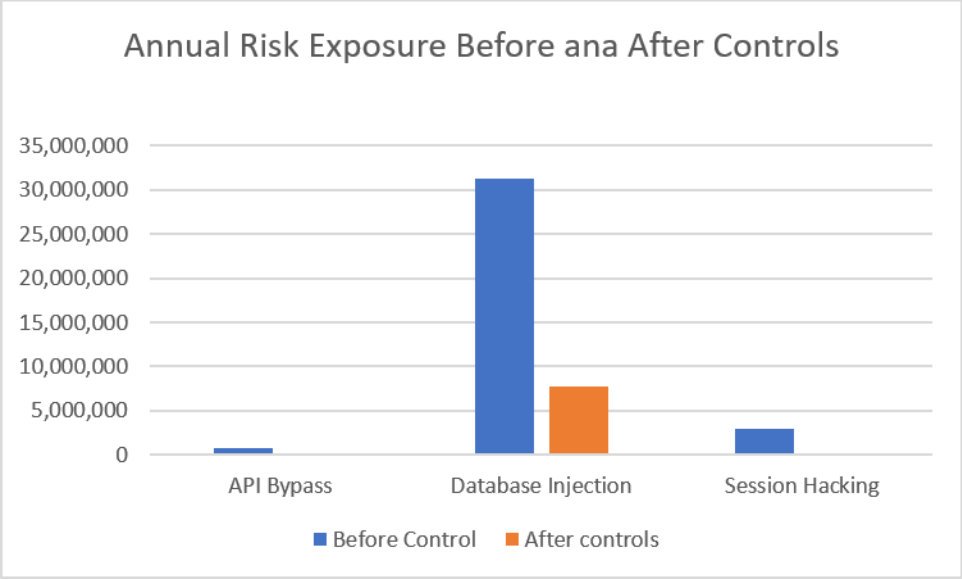
From: GRC Analyst

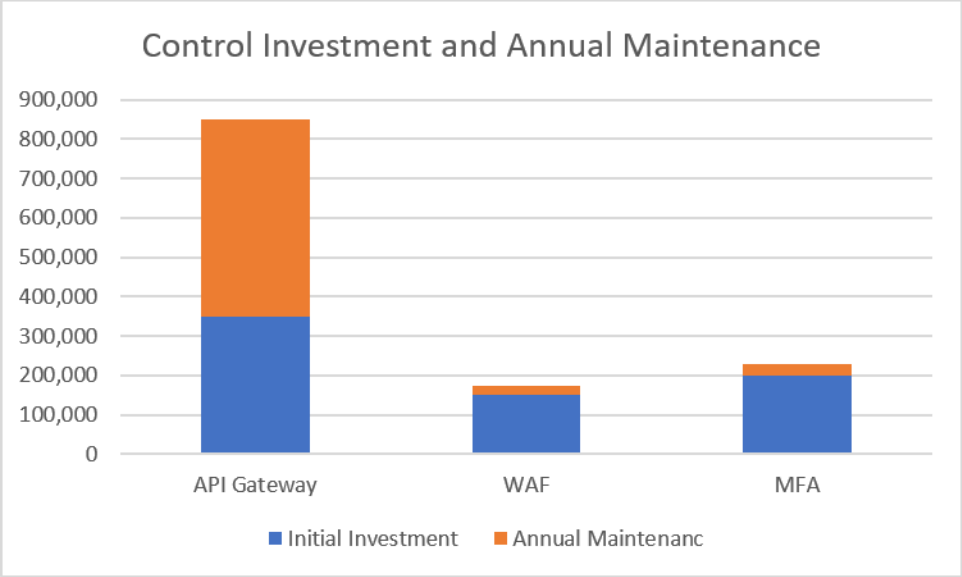
Date: 1ST November, 2026

Subject: Financial Risk Assessment and Recommended Controls for SecureMobile

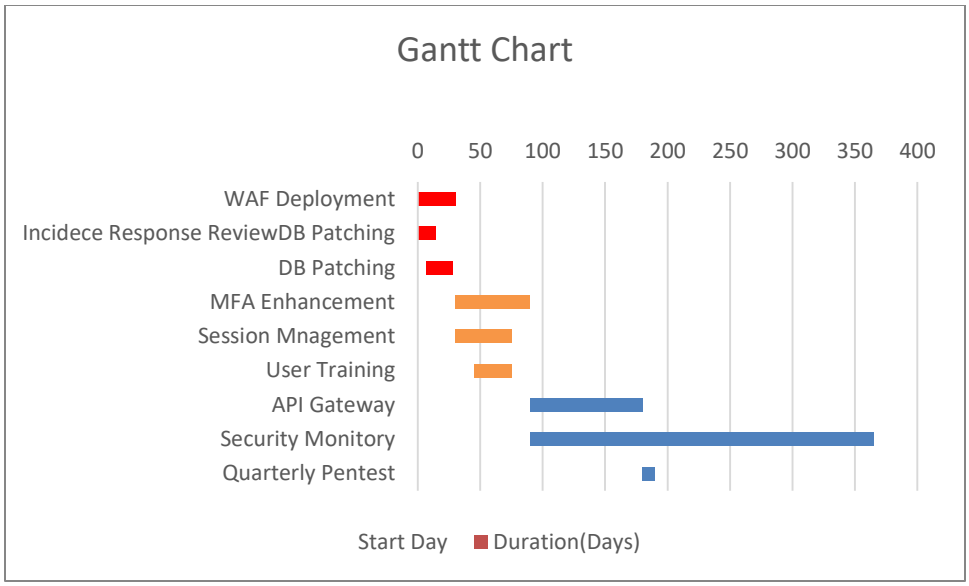
The assessment found three important vulnerabilities with a combined estimated annual loss exposure of $35,000,000. Database injection is the most significant risk, with an ALE of $31,250,000, followed by session hijacking at $3,000,000 and API authentication bypass at $750,000. The bank should prioritize the database injection vulnerability because it could expose personal and financial information belonging to customers.

I recommend implementing all three controls: the Advanced API Security Gateway, WAF and MFA enhancement. The combined initial investment is $700,000, with annual maintenance of $105,000. Based on the stated effectiveness rates, the estimated annual risk reduction is $26,962,500, leaving an estimated residual ALE of $8,037,500. This represents a reduction of 77.0% in the modelled annual exposure.





Task 7: Risk Treatment Timeline


Gantt Chart
0
50
100
150
200
250
300
350
400
WAF Deployment
Incidece Response ReviewDB Patching
DB Patching
MFA Enhancement
Session Mnagement
User Training
API Gateway
Security Monitory
Quarterly Pentest
Start Day
Duration(Days)